

Distinct Routes II

Time limit: 1.00 s

Memory limit: 512 MB

A game consists of n rooms and m teleporters. At the beginning of each day, you start in room 1 and you have to reach room n .

You can use each teleporter at most once during the game. You want to play the game for exactly k days. Every time you use any teleporter you have to pay one coin. What is the minimum number of coins you have to pay during k days if you play optimally?

Input

The first input line has three integers n , m and k : the number of rooms, the number of teleporters and the number of days you play the game. The rooms are numbered $1, 2, \dots, n$.

After this, there are m lines describing the teleporters. Each line has two integers a and b : there is a teleporter from room a to room b .

There are no two teleporters whose starting and ending room are the same.

Output

First print one integer: the minimum number of coins you have to pay if you play optimally. Then, print k route descriptions according to the example. You can print any valid solution.

If it is not possible to play the game for k days, print only -1.

Constraints

- $2 \leq n \leq 500$
- $1 \leq m \leq 1000$
- $1 \leq k \leq n - 1$
- $1 \leq a, b \leq n$

Example

Input	Output
8 10 2 1 2 1 3 2 5 2 4 3 5 3 6 4 8 5 8 6 7 7 8	6 4 1 2 4 8 4 1 3 5 8